

SpineMRI Foraminal Stenosis Grader

Clinical AI Report · Generated 14 July 2026, 11:35

Patient ID	Not specified	Model	EfficientNet-B0
Date	14 July 2026	Macro F1	0.607
Dataset	282 test samples	Accuracy	72.3%

Predicted Grade

Normal — Grade 0

Confidence: 95.5%

No significant foraminal narrowing detected. Perineural fat is preserved and the nerve root appears uncompressed.

Class Probability Distribution

Grade	Class	Probability	Confidence
Grade 0	Normal (Grade 0)	95.5%	██
Grade 1	Mild (Grade 1)	1.5%	██
Grade 2	Moderate (Grade 2)	2.0%	██
Grade 3	Severe (Grade 3)	1.0%	██

MRI Images



Original MRI crop (224×224)



Grad-CAM — warmer = higher activation

Grade Scale Reference

Grade	Severity	Clinical Description
Grade 0	Normal	No foraminal narrowing. Perineural fat intact. No nerve root compression.
Grade 1	Mild	Mild narrowing. Partial fat obliteration. Nerve root not yet deformed.
Grade 2	Moderate	Moderate compression. Complete fat obliteration. Nerve root contact present.
Grade 3	Severe	Severe compression. Nerve root deformation. Surgical consult may be needed.

DISCLAIMER: This report is generated by an AI research tool for educational and research purposes only. It does not constitute a medical diagnosis or substitute for evaluation by a qualified radiologist or physician.